

CITYCARE GENERAL HOSPITAL

ANTIBIOTIC POLICY STANDARD OPERATING PROCEDURE

Antimicrobial Stewardship, Prescribing Restrictions & Infection Management

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1. PURPOSE & SCOPE

Antimicrobial resistance (AMR) is one of the greatest threats to global public health. This Antibiotic Policy SOP establishes CityHospital's Antimicrobial Stewardship Programme (ASP), providing evidence-based guidance for the prescribing, monitoring, and review of all antimicrobial agents. Its goal is to optimise clinical outcomes, minimise adverse effects, and prevent the emergence and spread of antimicrobial resistance within the hospital and the wider community.

- Promote rational, evidence-based antibiotic prescribing across all clinical departments.
- Establish a tiered restriction system to preserve the efficacy of last-resort antibiotics.
- Standardise empirical therapy for common infections while rapidly directing therapy by culture results.
- Mandate the antibiotic start–review–stop cycle to prevent unnecessary prolonged courses.
- Monitor antibiotic consumption, resistance patterns, and clinical outcomes continuously.
- Comply with national antimicrobial stewardship framework requirements and accreditation standards.

WARNING: Broad-spectrum antibiotics including carbapenems, colistin, linezolid, and tigecycline are RESTRICTED drugs. Unauthorised prescribing of restricted antibiotics will be reversed by the antimicrobial stewardship team and reported as a clinical governance event.

2. DEFINITIONS & ABBREVIATIONS

Term / Abbreviation	Definition
ASP	Antimicrobial Stewardship Programme — the coordinated hospital programme to optimise antibiotic use.
AMR	Antimicrobial Resistance — resistance of microorganisms to antimicrobial medicines they were previously sensitive to.
MRSA	Methicillin-Resistant Staphylococcus aureus — major hospital-acquired pathogen.
ESBL	Extended-Spectrum Beta-Lactamase — enzyme-producing organisms resistant to most penicillins and cephalosporins.
CRE	Carbapenem-Resistant Enterobacteriaceae — organisms resistant to carbapenem antibiotics.
C. difficile	Clostridioides difficile — antibiotic-associated diarrhoea organism; risk increases with broad-spectrum antibiotic use.
MIC	Minimum Inhibitory Concentration — the lowest antibiotic concentration that prevents visible microbial growth.
IV-to-Oral Switch	Clinical programme to convert intravenous antibiotic to an equivalent oral preparation when clinically appropriate.
Empirical Therapy	Antibiotic treatment started before microbiological results are available, based on likely pathogens.
Targeted Therapy	Antibiotic therapy directed by confirmed culture and sensitivity results.
De-escalation	Narrowing or stopping antibiotic therapy once culture results confirm a more targeted agent is appropriate.
PK/PD	Pharmacokinetics and Pharmacodynamics — the relationship between drug exposure and antimicrobial effect.
OPAT	Outpatient Parenteral Antibiotic Therapy — IV antibiotics administered in a community or outpatient setting.

3. ROLES & RESPONSIBILITIES — ANTIMICROBIAL STEWARDSHIP TEAM

Role	Department	ASP Responsibilities
ASP Lead (Infectious Disease Physician)	Infectious Diseases	Programme leadership, restricted antibiotic authorisation, clinical advice
Clinical Microbiologist	Microbiology Lab	Interpreting cultures, AMR surveillance, advising on targeted therapy
ASP Pharmacist	Pharmacy	Daily antibiotic reviews, IV-oral switch, restriction enforcement, audits
Infection Control Nurse	IPC	Outbreak management, AMR organism isolation, HAI surveillance
Ward Physicians / Prescribers	All Specialties	Prescribe per policy; document indication; attend antibiotic review
Microbiology Lab Technician	Microbiology	Process specimens; report sensitivities; alert critical results
IT / EMR Team	Information Systems	Maintain antibiotic alerts, allergy records, stop dates in EMR

4. ANTIBIOTIC CLASSIFICATION & RESTRICTION TIERS

All antibiotics are classified into one of four access tiers aligned with the WHO AWaRe (Access, Watch, Reserve) framework, adapted for the local resistance epidemiology of CityHospital.

Tier	WHO AWaRe Category	Description	Prescribing Authority
1 — Access	ACCESS	First and second-line agents for common infections; low resistance risk. e.g., amoxicillin, trimethoprim, metronidazole, doxycycline.	All registered doctors
2 — Watch	WATCH	Higher resistance potential; used only when Access antibiotics are inappropriate. e.g., co-amoxiclav, cephalosporins, fluoroquinolones, clindamycin.	Specialist MO or above; indication documented
3 — Reserve	RESERVE	Last-resort antibiotics for MDR organisms; significant resistance risk. e.g., carbapenems, vancomycin, linezolid, colistin, tigecycline.	ASP Lead or Microbiologist authorisation required
4 — Controlled	HOSPITAL-SP ECIFIC	Novel or experimental antimicrobials; compassionate use only.	CMO and ASP Lead dual authorisation; PTC notification

4.1 Tier 3 Reserve Antibiotic Authorisation Process

Step 1. Prescribing physician contacts the ASP pharmacist or ASP Lead directly (phone or pager).

Step 2. Clinical case presented: diagnosis, microbiology results, antibiotics already tried, clinical urgency.

Step 3. ASP Lead or Microbiologist provides verbal authorisation; authorisation code entered in the EMR prescription field.

Step 4. Formal written authorisation completed in EMR within 4 hours of verbal approval.

Step 5. Prescribing duration specified at initiation; maximum 5-day course without re-authorisation.

Step 6. Review date entered in EMR; ASP pharmacist performs clinical review on day 3 of therapy.

5. EMPIRICAL ANTIBIOTIC THERAPY GUIDELINES

Empirical antibiotic selection must be based on the CityHospital local empirical guidelines, updated annually using the hospital antibiogram. The following table provides first-line empirical guidance for the most common infectious presentations. Always check for allergies before prescribing.

Infection / Syndrome	Likely Pathogen(s)	First-Line Empirical Treatment	Duration
Community pneumonia (non-severe)	S. pneumoniae, H. influenzae	Amoxicillin 500 mg TDS PO	5 days
Community pneumonia (severe/ICU)	As above + Legionella, Staph.	Co-amoxiclav IV + Clarithromycin IV	7–10 days
Urinary tract infection (uncomplicated)	E. coli, Klebsiella	Nitrofurantoin 100 mg BD PO	3 days (women)
Urinary tract infection (complicated/pyelonephritis)	E. coli, Gram negatives	Co-amoxiclav IV or Gentamicin IV	7–14 days
Skin & soft tissue (non-purulent)	Streptococcus, Staph aureus	Flucloxacillin 500 mg QDS PO/IV	5–7 days
Sepsis (source unknown)	Gram pos + neg, anaerobes	Piperacillin-tazobactam IV (Tier 2; review at 48 hrs)	Source-dependent
Intra-abdominal infection	E. coli, anaerobes, Klebsiella	Co-amoxiclav IV + Metronidazole IV	4–7 days post-source control
Meningitis (bacterial, adult)	N. meningitidis, S. pneumoniae	Ceftriaxone 2g BD IV + Dexamethasone	10–14 days
C. difficile infection (mild-moderate)	Clostridioides difficile	Metronidazole 400 mg TDS PO	10 days
C. difficile infection (severe)	Clostridioides difficile	Vancomycin 125 mg QDS PO (Tier 3)	10–14 days

NOTE: These guidelines are updated annually. Always check the current antibiogram on the hospital intranet before prescribing. Local resistance patterns may require deviation from first-line recommendations.

6. DE-ESCALATION, IV-TO-ORAL SWITCH & STOP PROTOCOLS

6.1 Antibiotic De-escalation

- Review all empirical antibiotic prescriptions at 48–72 hours using available clinical and microbiological data.
- De-escalate from broad-spectrum to targeted, narrower-spectrum agent as soon as culture sensitivities confirm the causative organism.
- If cultures are negative and patient has clinically improved: reassess antibiotic need; stop if no longer indicated.
- Document de-escalation decision and rationale in EMR at the 48-hour review.
- The ASP pharmacist prompts all prescribers regarding antibiotic review at 48–72 hours via EMR alert.

6.2 IV-to-Oral Switch Criteria

IV antibiotics should be switched to oral equivalent as soon as the patient meets ALL of the following criteria:

- Temperature below 38°C for at least 24 hours and clinical signs of infection improving.
- WBC trending towards normal range; CRP falling.
- Patient tolerating oral intake; no vomiting or significant gastrointestinal malabsorption.
- No condition requiring IV route specifically (e.g., endocarditis, meningitis, osteomyelitis in early phase).
- Oral formulation available with adequate bioavailability for the chosen agent.

6.3 Antibiotic Stop Protocol — 'Start Smart, Then Focus'

- Step 1.** Every antibiotic prescription must have a documented clinical indication and a planned stop date or review date entered in the EMR at the time of prescribing.
- Step 2.** Default stop date: 5 days for most community-acquired infections unless clinical evidence supports longer duration.
- Step 3.** ASP pharmacist reviews all antibiotic prescriptions exceeding 7 days; escalates to ASP Lead for clinical review.
- Step 4.** Antibiotic courses exceeding 14 days require documented ASP Lead review and justification entered in EMR.
- Step 5.** Automatic stop-date alerts sent to prescriber via EMR; prescription must be actively renewed with documentation or will expire.

7. CULTURE, SENSITIVITY & TARGETED THERAPY WORKFLOW

7.1 Pre-Treatment Specimen Collection

- Blood cultures: minimum 2 sets (aerobic + anaerobic) collected from 2 separate venepuncture sites before antibiotic initiation in all suspected bacteraemia cases.
- Urine culture, wound swab, sputum, or CSF collected as clinically indicated before first antibiotic dose where time permits.
- In septic shock (Level 1 emergency): culture collection must not delay antibiotic initiation beyond 1 hour.
- Culture specimen bottles labelled with patient MRN, site, date, and time; request form includes current antibiotics and clinical indication.

7.2 Microbiology Result Action Pathway

Step 1. Preliminary Gram stain result available: 1–2 hours. Microbiologist contacts clinician for clinically significant results.

Step 2. Formal sensitivity result available: 24–72 hours. Positive blood culture result telephoned immediately to treating team.

Step 3. Prescriber reviews sensitivities: de-escalate, escalate, or continue empirical therapy based on organism, MIC, and clinical response.

Step 4. ASP pharmacist reviews all positive culture results daily; recommends targeted therapy changes to prescriber.

Step 5. Prescriber updates antibiotic prescription in EMR; rationale for any deviation from sensitivity-guided recommendation documented.

Step 6. Final sensitivities filed in EMR; allergy field updated if new sensitivities or resistance patterns identified.

7.3 Multi-Drug Resistant Organism (MDRO) Pathway

- Confirmed MRSA, ESBL, CRE, or VRE: Infection Control Nurse and ASP Lead notified within 2 hours of lab result.
- Patient placed in contact isolation immediately; IPC precautions implemented.
- ASP Lead determines targeted therapy; treatment plan entered in EMR and communicated to MDT.
- MDRO colonisation or infection reported to the national AMR surveillance database.
- Patient and family counselled on the organism and infection control precautions required.

8. SPECIAL POPULATIONS & HIGH-RISK SCENARIOS

8.1 Renal Impairment

- Renal function (eGFR/CrCl) documented before prescribing any renally cleared antibiotic.
- Dose adjustment applied for: aminoglycosides, vancomycin, beta-lactams, carbapenems, and fluoroquinolones as per pharmacy renal dosing guide.
- Aminoglycosides in renal impairment: once-daily extended-interval dosing protocol; trough levels monitored.
- Vancomycin in renal impairment: pharmacist-led dose individualisation using AUC/MIC-guided dosing.

8.2 Hepatic Impairment

- Hepatically metabolised antibiotics (metronidazole, clindamycin, linezolid, rifampicin) require dose reduction in severe liver disease.
- Child-Pugh score documented; pharmacist review mandatory for Child-Pugh B or C patients.

8.3 Pregnancy & Breastfeeding

- Safe antibiotics in pregnancy: amoxicillin, ampicillin, cephalosporins, erythromycin, and nitrofurantoin (avoid near term).
- Strictly contraindicated in pregnancy: fluoroquinolones, tetracyclines, co-trimoxazole (first trimester), metronidazole (first trimester).
- All antibiotic prescriptions in pregnant patients require a pharmacist review; ASP Lead consulted for complex cases.

8.4 Penicillin Allergy Assessment

- Penicillin allergy documented in EMR at registration; nature of reaction recorded (rash, anaphylaxis, intolerance).
- Formal allergy de-labelling programme: structured allergy history taken by pharmacist or allergist for all patients labelled 'penicillin allergic'.
- True IgE-mediated allergy (anaphylaxis): avoid all penicillins and carbapenems; use alternative classes per ASP advice.
- Low-risk allergy (rash only): cross-reactivity with cephalosporins less than 2%; can use with monitoring and documented risk-benefit discussion.

8.5 Neutropenic Fever / Immunocompromised Patients

- Neutropenic fever (ANC below $0.5 \times 10^9/L$ and temperature above $38^\circ C$): broad-spectrum IV antibiotic within 30 minutes of recognition (Tier 2 escalation pre-authorised for this indication).
- Standard empirical protocol: piperacillin-tazobactam IV; escalate to meropenem if haemodynamically unstable or prior ESBL isolation.
- Antifungal prophylaxis or empirical treatment per oncology/haematology protocol if fever persists beyond 72 hours.

9. ANTIBIOTIC PROPHYLAXIS GUIDELINES

Surgical antibiotic prophylaxis (SAP) reduces the risk of surgical site infection when administered correctly. Prophylaxis outside these guidelines is not permitted without documented ASP authorisation.

9.1 Principles of Surgical Antibiotic Prophylaxis

- Administer within 60 minutes before surgical incision (30 minutes for vancomycin and fluoroquinolones due to infusion time).
- Single dose is sufficient for most procedures; repeat dose if operation exceeds 4 hours or blood loss exceeds 1500 ml.
- Prophylaxis should be discontinued within 24 hours of surgery — continuation beyond this is treatment, not prophylaxis.
- Select prophylactic antibiotic based on likely wound contaminants, not the broadest-spectrum agent available.

9.2 Surgical Prophylaxis Quick Reference

Surgical Procedure	Recommended Agent	Dose	Stop
General surgery (clean-contaminated)	Co-amoxiclav IV	1.2g at induction	Single dose
Colorectal surgery	Co-amoxiclav IV + Metronidazole IV	1.2g + 500mg at induction	Single dose
Orthopaedic (joint replacement)	Cefuroxime IV	1.5g at induction; repeat at 8h	Within 24 hours
Cardiac / vascular surgery	Cefuroxime IV	1.5g at induction	Within 24 hours
Penicillin allergy (all categories)	Vancomycin IV + Gentamicin IV	As per weight; infuse over 60 min	Single dose
Urological procedures (instrumentation)	Gentamicin IV	3–5 mg/kg at induction	Single dose
Neurosurgery (craniotomy)	Cefuroxime IV	1.5g at induction	Single dose

9.3 Medical Prophylaxis

- Endocarditis prophylaxis: restricted to high-risk patients (prosthetic valves, previous IE, congenital cyanotic heart disease) undergoing high-risk dental procedures. Amoxicillin 2g PO 30–60 min before procedure.
- Pneumocystis pneumonia (PCP) prophylaxis: co-trimoxazole 480 mg daily for all patients on immunosuppressant therapy with CD4 count below 200 cells/ μ l.
- Meningococcal contact prophylaxis: rifampicin or ciprofloxacin as per Public Health guidance; ASP Lead coordinates.

10. STEWARDSHIP AUDIT, AMR SURVEILLANCE & BEST PRACTICES

10.1 ASP Audit Framework

Audit Indicator	Target	Frequency
48-72hr antibiotic review documentation compliance	100%	Weekly
Empirical antibiotic choice concordance with local guidelines	Greater than 90%	Monthly
IV-to-oral switch conversion rate (eligible patients)	Greater than 80%	Monthly
Surgical prophylaxis administered within 60 min of incision	100%	Monthly per theatre list
Prophylaxis discontinued within 24 hours	100%	Monthly
Reserve antibiotic prescriptions with ASP authorisation code	100%	Weekly
Antibiotic prescription with documented indication	100%	Monthly random sample

10.2 AMR Surveillance

- Hospital antibiogram published quarterly, reporting organism susceptibility patterns for the top 10 pathogens from clinical cultures.
- Point prevalence survey (PPS) of antibiotic use conducted annually in alignment with ECDC methodology.
- HAI rates (MRSA bacteraemia, *C. difficile* infection) reported monthly to the infection control committee and clinical governance board.
- AMR data submitted to the national surveillance network as mandated by public health legislation.

10.3 Education & Best Practices

- Annual mandatory antimicrobial stewardship training for all prescribing doctors, pharmacists, and nurses.
- ASP induction training included in all new doctor and pharmacist orientation programmes.
- Antibiotic clinical guidelines published on the hospital intranet; quick-reference guide laminated and displayed at every ward prescribing station.
- Regular ASP Grand Rounds: quarterly case-based learning sessions on complex antibiotic cases and resistance trends.
- Prescriber feedback: individual-level antibiotic prescribing data shared confidentially with each prescriber quarterly.
- Patient-facing information: leaflets explaining why antibiotics are not prescribed for viral infections distributed in OPD waiting areas.

NOTE: CityHospital targets: MRSA bacteraemia rate below 0.5 per 1000 bed-days, *C. difficile* rate below 1.0 per 10,000 bed-days. These are reviewed monthly by the Infection Control Committee.